

Education

- 2022–Present **PhD., School of Computer Science, Carnegie Mellon University**, Pittsburgh, PA, US.
Societal Computing, Software and Societal Systems Department
Advisors: Yuvraj Agarwal, Mayank Goel.
- 2018–2021 **B.S., Department of Computer Science, Purdue University**, West Lafayette, IN, US.
Computer Science Honors, Minor in Mathematics
GPA: 3.97/4.00, Graduation with Highest Distinction.

Research Experience

- 2024–Present **Generalizable and Flexible On-Body IMU Sensing Systems**, [C.4],
Carnegie Mellon University, Pittsburgh, PA, US; Advisors: Yuvraj Agarwal, Mayank Goel.
 - Developed skeleton-guided motion feature transference using synthetic IMU on SMPL pose models.
 - Designed ML framework adaptive to devices placed on continuous human body spatial coordinates.
 - Achieved SOTA human pose estimation accuracy with on-body IMU with flexible placements.
- 2022–2024 **Multi-User Privacy Negotiation System for Shared IoT Devices**, [C.3],
Carnegie Mellon University, Pittsburgh, PA, US; Advisors: Yuvraj Agarwal, Mayank Goel.
 - Designed a negotiation system that strategically elicits user preference and proposes configurations.
 - Built a custom shared-space privacy profile model with survey responses from 198 mTurk workers.
 - Evaluated the negotiation system on 30 users and showed 83.3% overall outcome satisfaction.
- 2022–2024 **Privacy-Preserving Audio Featurization under Modern ASR**, [C.2, D.1],
Carnegie Mellon University, Pittsburgh, PA, US; Advisors: Yuvraj Agarwal, Mayank Goel.
 - Revealed speech content inference vulnerability on 4 prior privacy-oriented featurization approaches.
 - Presented an ML-based speech filter detects and removes speech on resource-constrained devices.
- 2019–2021 **Semantic Network of Kitchen Things for Cooking Assistance**, [J.1],
Purdue University, West Lafayette, IN, US; Advisors: Richard M. Voyles.
 - Designed appliance-domain and cooking-domain ontology to reason cooking states and context
 - Propotyped mobile app connecting with sensor-equipped smart food containers with BLE

Industrial Experience

- 2025 **Samsung Research America**, *Research Intern*, Plano, TX
May–Aug Research and development on next-generation connectivity and sensing technology.
- 2024 **Procter and Gamble**, *Research & Development Intern*, Mason, OH
May–Aug Data science in real-time in-context and privacy-preserving consumer understanding.
- 2022 **Trendalyze Inc.**, *Data Science Intern*, Newark, NJ (Remote)
Feb–May Electrocardiogram (ECG) timeseries analysis on QRS complex and R-peak detection.

Teaching Experience

- Fall 2025 **17720: Machine Learning and Sensing for Healthcare**, Graduate Teaching Assistant,
Carnegie Mellon University, Pittsburgh, PA, US.
Topics: sensor fusion; machine learning; acoustic ranging, beamforming; sensing for health .
- 2024 Spring **17722: Building User-Focused Sensing Systems**, Graduate Teaching Assistant,
Carnegie Mellon University, Pittsburgh, PA, US.
Topics: signal processing and ML with IMU, audio, vision; research review on sensing .

2021 Fall **CS250: Computer Architecture**, Undergraduate Teaching Assistant, Purdue University.
2021 Fall **STAT290: The Data Mine II**, Undergraduate Teaching Assistant, Purdue University.
2021 Spring **CS390: Neural Image Processing**, Undergraduate Teaching Assistant, Purdue University.

Fellowships & Awards

2024 Aug **CMLH Fellowship in Digital Health Innovation**, Carnegie Mellon University
2023 Oct **Sansom Graduate Fellowship**, Carnegie Mellon University
2021 Dec **Graduation with Highest Distinction**, Purdue University
2021 Apr **College of Science Alumni Summer Research Fellowship**, Purdue University

Skills and Competencies

Sensing and Signals:	IMU, Audio, Vision, mmWave, PIR; Mobile, Wearable, Ambient Sensors
Machine Learning:	Computer Vision, NLP, Multimodal ML; PyTorch, TensorFlow
LLM and Agents:	RAG, tool-augmented LLMs, Workflow Orchestration, Data Synthesis
Software Development:	Python, Java, C/C++, Javascript, HTML, Vue, React, Android, iOS
Research Methods:	System Prototyping, Human-Centered Design, User Studies

Presentations & Invited Talks

2024 Sept **CyLab Partner Conference** *Bring Privacy To The Table: Interactive Negotiation for Privacy Settings of Shared Sensing Devices*, Pittsburgh, PA, US.

Academic Services

2024-2025 Paper Reviewer: IMWUT, UIST; IEEE Transactions on Mobile Computing
Student Volunteer: UIST'24

Publications

Refereed Conference Articles

- [C.4] **Haozhe Zhou**, Riku Arakawa Yuvraj Agarwal, Mayank Goel. 2025. IMUCoCo: Enabling Flexible On-Body IMU Placement for Human Pose Estimation and Activity Recognition. *The ACM Symposium on User Interface Software and Technology* (UIST '25)
- [C.3] **Haozhe Zhou**, Mayank Goel, Yuvraj Agarwal. 2024. Bring Privacy To The Table: Interactive Negotiation for Privacy Settings of Shared Sensing Devices. *CHI Conference on Human Factors in Computing Systems* (CHI '24)
- [C.2] Sudershan Boovaraghavan, **Haozhe Zhou**, Mayank Goel, Yuvraj Agarwal. 2024. Kirigami: Lightweight Speech Filtering for Privacy-Preserving Activity Recognition using Audio. *ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (UbiComp '24)
- [C.1] Upinder Kaur, **Haozhe Zhou**, Xiaxin Shen, Byung-Cheol Min, and Richard M. Voyles. 2021. RoboMal: Malware Detection for Robot Network Systems. *IEEE International Conference on Robotic Computing*. (IRC '21)

Refereed Journal Articles

- [J.1] Sangjun Eom, **Haozhe Zhou**, Upinder Kaur, Richard M. Voyles, David Kusuma. 2022. TupperwareEarth: Bringing Intelligent User Assistance to the "Internet of Kitchen Things". *IEEE Internet of Things Journal*.

Demos and Posters

- [D.1] **Haozhe Zhou**, Sudershan Boovaraghavan, Mayank Goel, Yuvraj Agarwal. 2024. On-Device Speech Filtering for Privacy-Preserving Acoustic Activity Recognition. *The 30th Annual International Conference on Mobile Computing and Networking*. (MobiCom '24)